ERIC JIN

Senior AI Engineer | Staff Engineer roles
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Summary

Senior Full-Stack Engineer and AI Systems Developer with over **14 years of experience** designing and building scalable SaaS platforms, cloud-native infrastructure, and AI-powered applications. Proven expertise in developing **LLM-driven systems, intelligent automation workflows, and data-driven platforms** using modern technologies including **Node.js, Python, React, and TypeScript**.

Experienced in integrating advanced AI capabilities using platforms such as OpenAI API, implementing **retrieval-augmented generation (RAG), semantic search, and AI-powered analytics systems**. Strong background in backend architecture, distributed systems, and API design, with production experience deploying cloud applications across Amazon Web Services, Google Cloud Platform, and Microsoft Azure.

Skilled in designing **high-performance APIs, microservices architectures, and scalable data pipelines**, with hands-on expertise in containerized environments using Docker and Kubernetes. Adept at leading end-to-end system development, from architecture design to production deployment, while delivering reliable and maintainable software solutions for enterprise and startup environments.

Passionate about building **AI-driven products that combine machine intelligence with scalable cloud infrastructure** to deliver real-world business impact.

Skills

AI & Machine Learning:
Large Language Models (LLMs), Generative AI Systems, Prompt Engineering, AI Workflow Automation, Retrieval Augmented Generation (RAG), Embeddings, Semantic Search, AI Agents and Tool Integration, Model Evaluation and Optimization

Frameworks & Libraries:
OpenAI API, LangChain, LlamaIndex, Hugging Face Transformers, PyTorch, TensorFlow, Scikit-learn

Backend & API Development:
Node.js, Express.js, NestJS, Python (FastAPI, Django, Flask), REST APIs, GraphQL APIs, Microservices Architecture, Authentication (OAuth, JWT), API Gateway and Service Integration

Frontend Development:
React, TypeScript, JavaScript (ES6+), Next.js, Vite, HTML5, CSS3, Tailwind CSS, Responsive UI Development, Data Visualization (Recharts, Chart.js)

Cloud & DevOps:
AWS(EC2, S3, Lambda, RDS, IAM, SQS), Microsoft Azure, Google Cloud Platform, Docker, Kubernetes, CI/CD Pipelines, Infrastructure as Code

Data Engineering & Databases:
PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, Data Pipelines and ETL Workflows, Vector Databases and Semantic Search, Pinecone, Weaviate, Milvus

Software Architecture:
Scalable SaaS Architecture, Distributed Systems Design, Cloud-Native Application Development, High-Performance API Systems, Event-Driven Architecture

Development Tools:
Git, GitHub, GitLab, Linux / Unix Environments, Agile / Scrum Development, Technical Documentation and Code Reviews

Work History

Webasen
Remote

Artificial Intelligence Engineer | Senior Full-stack Developer | Part-Time 09/2025 - Present

- Design and develop **AI-powered web applications and automation systems** integrating Large Language Models using the OpenAI API to enable intelligent content generation, data analysis, and workflow automation.
- Build scalable **backend services and APIs** using Node.js and Python, implementing microservices architectures for high-performance data processing and AI-driven features.
- Implement **Retrieval-Augmented Generation (RAG)** pipelines using vector embeddings and semantic search to improve accuracy of AI-generated responses and enterprise knowledge retrieval.
- Develop modern **frontend interfaces** using React and TypeScript to deliver responsive, data-driven user experiences integrated with AI services.
- Architect and deploy **cloud-native applications** across Amazon Web Services and Google Cloud Platform, ensuring scalability, reliability, and cost-efficient infrastructure.
- **Tech Stack:** Python, Node.js, React, TypeScript, OpenAI API, LangChain, PostgreSQL, Docker, Kubernetes, AWS, GCP.

LocalSpark
Remote

Full-stack Developer | AI Engineer | Project Manager | Full-Time 09/2023 - 01/2026

- Led the design and development of **AI-powered SaaS platforms and analytics systems**, integrating Large Language Models through the OpenAI API to enable intelligent automation, predictive insights, and data-driven decision making.
- Architected and implemented scalable **full-stack web applications** using Node.js, Python, React, and TypeScript, delivering high-performance APIs and modern user interfaces for enterprise clients.
- Managed and mentored cross-functional engineering teams in an **Agile development environment**, coordinating project planning, technical design, and product delivery.
- Developed scalable **REST and GraphQL APIs** supporting large-scale data processing and real-time analytics systems.
- Containerized and deployed applications using Docker and automated deployments through CI/CD pipelines for faster and reliable releases.
- Collaborated closely with product stakeholders and business teams to translate **technical architecture into scalable SaaS solutions** used by multiple organizations.
- **Tech Stack:** Node.js, Python, React, TypeScript, OpenAI API, PostgreSQL, MongoDB, Docker, AWS, Azure, CI/CD.

BussinessInsuranceUSA
Remote

Full-stack Developer | Contract 01/2022 - 07/2023

- Developed and maintained scalable **full-stack web applications** supporting online insurance services, improving platform performance and user experience.
- Built robust **backend APIs and microservices** using Node.js and Python to handle client data processing, authentication, and business logic for insurance-related workflows.
- Designed and implemented responsive **frontend interfaces** using React and modern JavaScript frameworks to deliver seamless and intuitive user experiences.
- Integrated third-party services and APIs to support **payment processing, document management, and customer data systems**.
- Optimized database queries and backend logic using PostgreSQL and MongoDB to improve application performance and data reliability.
- Implemented secure authentication systems and role-based access control to protect sensitive customer and policy information.
- Deployed and maintained applications on cloud infrastructure including Amazon Web Services, ensuring reliable production environments.
- **Tech Stack:** Node.js, Python, React, JavaScript, PostgreSQL, MongoDB, AWS, REST APIs.

Work History

LeadAction
Remote

Full-stack Developer | Full-Time
09/2017 - 12/2021

- Developed and maintained scalable **SaaS platforms and web applications** supporting marketing automation and customer engagement systems used by businesses to manage campaigns and communication workflows.
- Designed and implemented **high-performance backend services and RESTful APIs** using Node.js and Python, enabling reliable data processing and integration with external services.
- Built modern and responsive **frontend applications** using React and JavaScript, delivering intuitive dashboards and user interfaces for campaign management and analytics.
- Implemented **email and SMS automation workflows**, supporting large-scale messaging infrastructure and customer engagement tools.
- Designed and optimized **database schemas and queries** using PostgreSQL and MySQL to support high-volume transactional and analytics workloads.
- Integrated third-party APIs for **payment systems, marketing platforms, and communication services**, expanding the platform's automation capabilities.
- **Tech Stack:** Node.js, React, JavaScript, PostgreSQL, MySQL, Redis, REST APIs, AWS.

Alpaca
Hybrid

Intern | Full-stack Developer | Contract
08/2012 - 06/2017

- Started as a **software engineering intern** and progressed into a **Full-Stack Developer role**, contributing to the development of internal tools and web-based applications.
- Developed responsive **frontend interfaces** using JavaScript, HTML5, and CSS, improving usability and accessibility across desktop and mobile devices.
- Participated in the development of **database-driven applications**, writing optimized queries and maintaining relational databases such as MySQL and PostgreSQL.
- Contributed to the integration of third-party APIs and external services to extend application functionality.
- Collaborated with senior engineers and designers to deliver product features, fix bugs, and improve overall platform performance.
- Gained hands-on experience with **software development lifecycle practices**, version control systems, and Agile development workflows.
- **Tech Stack:** JavaScript, Node.js, PHP, HTML, CSS, MySQL, REST APIs, Git.

Education

DeepLearning.AI
Online

Generative AI with Large Language Models
2024

Stanford University(Online Program)
Online

Machine Learning Specialization
2021

Kanagawa University
Kanagawa, Japan

Bachelor's degree in Computer Science
04/2010 - 03/2014

Certification

Generative AI with Large Language Models | DeepLearning.AI

Google Professional Machine Learning Engineer | Google Cloud

AWS Certified Solutions Architect – Associate | Amazon Web Services

Certified Kubernetes Administrator (CKA) | Cloud Native Computing Foundation

Projects

https://webasen.com/

Developed an AI-powered platform designed to automate business workflows and enhance decision-making through intelligent data processing and natural language capabilities. The platform integrates large language models via the OpenAI API to enable advanced features such as automated content generation, semantic search, and conversational AI tools.

https://www.localspark.ai/

Designed and implemented an AI-driven marketing analytics and communication platform for small businesses. Built with Laravel and modern frontend tooling, the system integrates OpenAI models to power intelligent campaign insights, automated content suggestions, lead analysis, and performance reporting. Implemented structured prompt workflows, response validation, and data-driven automation pipelines to enhance accuracy and scalability.

https://leadaction.com/

Contributed to the development of a scalable SaaS platform designed to help businesses automate marketing campaigns, manage customer engagement, and analyze campaign performance through data-driven dashboards. The platform provided tools for managing email and SMS campaigns, customer segmentation, and automated communication workflows.

https://www.businessinsuranceusa.com/

Contributed to the modernization of the insurance quoting platform, integrating intelligent form logic and backend optimizations. Enhanced workflow automation and implemented structured API communication layers designed to support scalable, data-driven decision logic and future AI-based underwriting enhancements.

https://wellnessentially.com/

Developed a MERN-stack mobile and web platform focused on employee well-being and productivity. Designed backend APIs and data pipelines structured to support behavioral analytics and future AI-powered recommendation systems. Implemented scalable architecture on Azure to handle real-time usage tracking and performance insights.

https://gardenkitchen.us/

Built a responsive restaurant platform featuring online ordering and reservations. Architected lightweight backend APIs and data tracking mechanisms to support automation, structured analytics collection, and potential AI-driven customer engagement enhancements.

https://installeranalytics.com/

Contributed to a SaaS analytics platform for Windows installers. Designed and enhanced reporting systems, backend microservices, and data processing pipelines. Implemented scalable architecture optimized for real-time analytics and structured data aggregation suitable for advanced AI-based insights and anomaly detection.

https://www.metamarketing.com/

Led development of a marketing automation suite integrating CRM data, analytics dashboards, and campaign management tools. Designed modular backend services using Node.js and Django, implemented scalable AWS-based infrastructure, and integrated AI-driven analytics workflows to automate insights, optimize campaign performance, and improve decision intelligence.